TITLE: Interfacing Relay & Buzzer with 8051 Microcontroller.

AIM:

To interface a Relay and Buzzer with the 8051 microcontroller and control their ON/OFF operation using an embedded C program.

After completion of this experiment, students will be able to

1) To control the relay and buzzer through 8051 I/O ports.
2) Program a 8051 microcontroller using Embedded C Language

THEORY:

The 8051 microcontroller operates at 5 V logic levels, but its I/O pins cannot directly drive devices such as relays because the relay coil requires comparatively higher current.

Therefore, a transistor driver circuit is used between the 8051 and the relay. When the corresponding port pin is made HIGH, the transistor turns ON and energizes the relay coil.

A flyback diode (1N4007) is connected across the relay coil to protect the transistor and microcontroller from the reverse voltage (back EMF) generated when the relay is switched OFF.

Similarly, a transistor driver is used to interface the buzzer with the 8051. The microcontroller controls the transistor through an I/O pin, thereby switching the buzzer ON or OFF.

Circuit Diagram:-

Procedure

1. Connect the 8051 microcontroller and driver circuits according to the circuit diagram.
2. Connect the relay to P1.0 through the BC547 transistor driver.
3. Connect the buzzer to P1.1 through the second transistor driver.
4. Connect the flyback diode across the relay coil with correct polarity.
5. Check all power and ground connections.
6. Compile the Embedded C program and generate the HEX file.
7. Load the HEX file into the 8051 microcontroller.
8. Switch ON the +5 V power supply.
9. Observe the relay and buzzer operation.
10. Verify the ON/OFF conditions according to the observation table.

Observation:-

Conclusions:

QUESTIONS:

1. What is the function of Relay & Buzzer?
2. How do you configure a 8051 Microcontroller i/p & o/p port pin?
3. List out the name of various SFR?
4. Write a Program to interface Relay & Buzzer to 8051 Microcontroller?